HOW TO SURVIVE ROBOTS VS HUMAN JOBS

Humanity is racing towards a future of fully automated luxury capitalism in a post-jobs world

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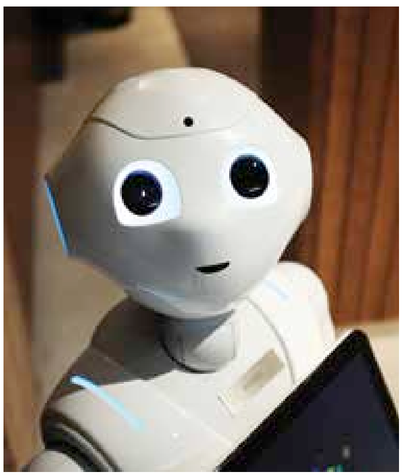
In the future, machines will replace humans in jobs. This is not controversial: it's what machines have done since well before the start of the industrial revolution. Petrol pump attendants were replaced by automated pumps, secretaries were replaced by Microsoft Office. This is what economists call the substitutive effect of automation: humans are substituted in jobs by machines.

THE COMPLEMENTARY EFFECT

From time to time, fears have been expressed that humans would run out of jobs entirely. I first wrote about this concern in 1980, and like many other people at the time, I under-estimated the resilience of the complementary effect of automation. To simplify a little, this means that when a human job is automated, the amount of wealth generated in the economy increases. That increases demand, which in turn increases employment. The new jobs that are created are done by humans because machines can not yet do everything that we can do for money.



The complementary effect probably won't last forever. The amount of compute power you can buy for $1,000 continues to increase exponentially: Moore's Law is evolving, not stopping. This means that in a decade, the machines we have will be 100 times as powerful as the ones we have today. In two decades the multiple will be 8,000, and in three decades, 1,000,000. There is no straight-line correlation between compute power and the ability of AI to carry out human tasks, but the two do travel together. People who airily dismiss fears about medium-term technological unemployment are committing the Reverse Luddite Fallacy. They are making four mistakes.

THE REVERSE LUDDITE FALLACY

First, they forget that we have seen technological unemployment before. In 1915 there were 22 million horses working in America, pulling vehicles and working on farms. The horse population of America today is two million. That is severe technological unemployment.

Second, they argue that because automation has not caused technological unemployment in the past, it will not do so in the future. The past is often a good guide to the future, but it is far from perfect: if it was, we would not be able to fly.

Third and fourth, they are thinking too short-term, and they are not taking into account the astonishing power of exponential growth. Machines will not be able to do everything humans can do for money in the next year, nor in the next decade. It is also true that